

Module 06: Learning — Smart Robots

Learning Objectives

1. Understand that some robots can **Learn**.
2. Learn about **Training** (Showing examples).
3. Know that robots learn from **Mistakes**.

Introduction: The Robot Student

Can a robot go to school? Sort of! Old robots only followed rules ("Turn Left"). If the map changed, they got stuck. New robots use **Machine Learning**. They practice!

- Try 1: Crash. (Bad!)
- Try 2: Crash. (Bad!)
- Try 3: Turn early. (Good!) The robot remembers what worked.

Key Concepts

1. Training Data (Examples)

How do you teach a robot what a cat looks like? You show it 1,000 pictures of cats. And 1,000 pictures of dogs. The robot finds the pattern (Cats have pointy ears). This is called **Training**.

2. Trial and Error

Robots learn by trying. A walking robot falls down 100 times. But every time it falls, its brain changes slightly. "Don't do *that* again." Eventually, it walks perfectly!

3. Updates

Your brain grows when you learn. A robot's code gets an **Update**. It gets smarter over time.

Activities

Activity 1: Hot or Cold?

Hide a "Treasure" (Toy). Your friend is the Robot. They move around. You give **Feedback**:

- "Colder..." (Wrong way).
- "Warmer..." (Getting closer).
- "HOT!" (Success). The robot uses feedback to find the goal!

Activity 2: Cat vs. Dog Sorter

Draw 3 Cats and 3 Dogs on cards. Mix them up. Ask a friend to sort them. Ask: "How did you know it was a cat?" (Pointy ears? Whiskers?) You are explaining your **Training Data**.

Summary

Learning makes robots smart. Instead of just following rules, they can figure things out by practicing and looking at examples.

References

- *The Wild Robot Escapes* by Peter Brown
- *Machine Learning for Kids* (Online tool)